

# TFPT and the Universalraum

## Source selection, reconstruction and accessible dynamics

Research consolidation

12 September 2026

### Abstract

We consolidate finite algebraic and analytic controls, four attributed research-task reports on state-to-generator reconstruction, and new exact tests of covariant edge operations. The latter yield an original-boundary signal under an explicitly additional event channel, but their fixed algebra is not one-dimensional. Reconstruction of a law from specified data is distinguished from deriving those data from a primitive source. All physical T1–T8 gates remain open.

## 1 Universalraum synthesis: source selection, reconstruction and accessible dynamics

**Scope and evidence, 12 September 2026.** This section consolidates the local primitive-transfer investigation and the latest two available answers from each of four ChatGPT research tasks. It is not an independent rerun of all external certificates, an exhaustive conversation archive, or a derivation of a theory of everything. All T1–T8 and TFPT.TOE.COMPLETE.01 remain open. The local checkout base is 66b91e40e245569f06ab440ead80f446c9be0ee5; additional local files must not be interpreted as published results.

The central distinction is between *reconstructing a generator from specified data* and *deriving those data and the elementary generator class from the primitive source*. Several finite reconstruction steps are now sharply formulated. The second, forward implication remains open. No RH, general efficient factoring, P versus NP, or Hylæan capability claim is promoted.

### 1.1 Typed objects and prior local results

The integral compiler commutant, its positive cone, the sixteen-real-mode quadratic candidate, the seventy-state interacting chain, the filled QWZ strip and the compact-U(1) rotor parent are different objects. In particular, the quadratic candidate has eight complex modes and a 256-dimensional full Fock space. The designated  $H_{70}$  is a fixed-four-particle sector of an open four-site, eight-mode model; it is not an unbounded winding rotor or the internal Dirac operator.

The compiler-order audit distinguishes a unique minimal family-inner order from its unique maximal overorder, with integral indices four and four. These solve different selection problems. The Hermitian  $2 \times 2$  formula

$$X = \begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix}, \quad \det X = t^2 - x^2 - y^2 - z^2$$

provides an exact Lorentz-cone representation, not a derived spacetime metric, action or quantum-gravity measure. Under the matched-center and Clock-normalizer hypotheses, the actual eight-complex-mode source excludes a unital  $M_2(\mathbb{C})$  identification carrying its whole Clock action. The active  $M_3$  corner accommodates two distinct slow frequencies; a closed autonomous  $M_2$  automorphism flow has only one positive Bohr frequency.

Other local controls delimit apparent simplifications. One-sided quaternion defects can be scalar, and Gram whitening can erase the relative rates. A unique Hilbert–Schmidt relational vector need not give a nontrivial physical reduced state. A four-point  $D_4$ -symmetric conservative real generator retains independent edge and diagonal weights. On the scalar circle, full rotation/reflection symmetry and differential order at most two imply  $aD^2 + b$ ; fourfold symmetry alone does not.

The direct positive point-evaluation perturbation of  $|D|$  is not a closed form in the audited setting; its positive Fourier-cutoff family has the free norm-resolvent limit. A negative renormalized coupling gives a nontrivial family with an unselected parameter. Under the direct rotor identification its selected vector need not have finite  $D^2$  energy. Adding a positive  $\eta D^2$  term changes the form domain and introduces an unselected  $\eta$ . The free Dirichlet–Neumann operator  $|D| = \sqrt{D^2}$  is an exact bridge, but not an identification of the two time generators.

Finally, for  $u^* = -u$  and  $u^2 = -I$ ,

$$(I + nu)/\sqrt{1 + n^2} = \exp(\arctan(n)u).$$

An integral index  $(1 + n^2)^4$  of the unnormalized lattice action is not entropy production of this unitary, nor a derived physical clock.

## 1.2 Results reported by the four parallel tasks

The following are *attributed task reports*; their original finite-field and rank certificates have not all been independently rerun in this consolidation. The archived excerpts retain exact task identifiers and retrieval limitations.

1. **Vollständige Lösung entwerfen.** For the designated  $H_{70}$ , reported irreducibility over  $\mathbb{Q}(\zeta_8)$  implies that linear zero conditions over this field annihilating the ground vector annihilate its entire 38-dimensional even reflection sector. Reductions at 449 and 977 are reported. This is a scoped obstruction to exact linear zero selection, not to all variational selection. The explicitly weighted two-state control

$$Q = \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix}, \quad \text{spec } Q = \{3 - \sqrt{5}, 3 + \sqrt{5}\}$$

has a simple positive minimum; changing relative weights changes its selected vector. A previous nearest-neighbor, traceless QWZ comparison reports rank 14 in a 15-parameter class, while allowing  $h + \epsilon h^3$  preserves occupied projectors and changes relative energies.

2. **TFPT und Universalraum Ideen.** A declared neutral, finite-range quartic extension contains 110 nonzero expressions spanning 66 independent real-symmetric directions, including the original 17. A real even eigenstate can constrain at most 37 transverse directions in the 38-dimensional even sector, leaving at least 28 nonconstant eigenstate-compatible directions. A separate thermal reader is reported to have kernel  $\text{span}\{I, H_{70}\}$  for every finite  $\beta > 0$ , using a rank-64 commutator certificate and thermal positivity. Synthetic thermal numerical reconstructions are diagnostics, not independent primitive-state derivations.
3. **Untersuche das Gesamtbild.** A different product-enlarged class is reported to have dimension 133 and 95 nonconstant compatible directions; it must not be confused with the 66-dimensional class. The signed equal-weight Laplacian on the designated 70 configurations and 220 edges has a unique frustrated minimum but its state admits no nonconstant generator in the old comparison family. This rejects that particular selector, not TFPT generally. Two designated states share  $C = I_8/2$  while their electric-form expectations differ, showing information loss in a one-body description.
4. **Thema vollständig lösen.** All seventy eigenvectors of the specified  $H_{70}$  are reported non-Slater, using Plücker-relation certificates in finite extensions at primes 347 and 23. Full two-particle

data determine its nondegenerate ground state. Within the original family, sixteen nonconstant expectations and a positive supporting-operator problem determine the generator direction, conditional on the reported rigidity. Neither the state data nor the choice of family have thereby been obtained from P1/P2.

The findings agree on a structural requirement: the source must distinguish elementary generators from products of already allowed operations. Closing the observable algebra under composition is not a derivation of a list of independently weighted primitive energy terms.

### 1.3 Reconstruction contracts that must remain distinct

For a pure eigenstate and a specified Hermitian family  $O_i$ , the covariance form obeys

$$c^\top \Gamma c = \|(K - \langle K \rangle)\psi\|^2, \quad K = \sum_i c_i O_i.$$

A one-dimensional nonconstant kernel identifies one generator direction, not the origin of the family or state. For a faithful mixed state,  $\text{Var}_\rho(K) = 0$  instead forces  $K$  to be scalar. Thermal reconstruction therefore needs its own stationarity-response contract. Adding separately measured eigenstate covariance matrices is not the same as mixing the eigenstates before computing one covariance matrix.

For a fixed- $N$  system with a Hamiltonian of at most two-body order, the complete two-particle reduced density matrix determines its energy expectation. If the ground state  $\rho_0$  is unique with gap  $\Delta > 0$ ,

$$H - E_0 I \succeq \Delta(I - \rho_0)$$

implies uniqueness of the positive state completion having the same two-particle data. More generally,

$$1 - \text{Tr}(\sigma \rho_0) \leq \frac{\text{Tr}(\sigma H) - E_0}{\Delta}.$$

This conditional argument is not a general uniqueness statement for arbitrary excited states. Likewise  $\log((I - C)C^{-1})$  reconstructs the quadratic quasifree one-body reader, not every interacting state with that covariance.

A different sufficient finite contract starts with a fully specified selfadjoint strictly positive Hilbert-space operator  $T$ , a simple largest eigenvalue  $\lambda_0$  and a positive step size  $\delta$ :

$$H = -\delta^{-1} \log(T/\lambda_0).$$

It yields a unique ground ray and energy differences. It does not determine  $T$  or  $\delta$  from symmetry or a common fixed vector. Computing  $T = e^{-\delta H_{70}}$  first and then inverting it is circular as an origin argument. A completely positive channel is not this type of operator.

### 1.4 Exact access results in the local quadratic candidate

For the pinned real matrices,

$$D = \alpha J + gB, \quad [O, J] = [O, B] = [J, B] = 0, \quad P = \frac{1}{6} \sum_{k=0}^5 O^k, \quad Q = I - P,$$

the real ranks are ten and six and  $PJQ = PBQ = 0$ . The original boundary cyclic space equals  $P$ . The induced Fock decomposition is  $32 \times 8$ ; with the unchanged split dynamics, finite adaptive boundary protocols depend only on the accessible marginal. Hidden occupation cannot be read out by such protocols, even if other initial correlations are allowed.

The hidden complex energies are  $\alpha - g, \alpha + \sqrt{3}g, \alpha - \sqrt{3}g$ . The neutral slow transitions have magnitudes  $(\sqrt{3} \pm 1)|g|$  and annihilate the empty vacuum at the audited point  $(\alpha, g) = (1, 1/8)$ .

A conditional added mixed field exposes all sixteen linear directions. An even cross bilinear can generate all 120 quadratic directions with the full even boundary algebra and free derivation; this does not prove arbitrary many-body control.

The invariant hidden operator algebra is  $M_2 \oplus M_2 \oplus \mathbb{C}^4$ . Its frequencies are  $0, \pm 2\alpha$ ; number conservation leaves only diagonal operators. This is a conditional restriction: the source has not proved that its Clock must be quotiented as gauge on every observable. The older  $\mu_4/\mathbb{Z}_8$  seam/GSO discussion concerns a different Clock and cannot be transferred without an intertwiner.

The existing  $B$  decomposes exactly into fifteen channel-edge matrices. Each of the six triple-to-pair edges, if separately available together with  $D$ , opens all sixteen real linear directions. A modified fixed weight  $B + xE_{14}$  also gives full access for every real  $x \neq 0$ : an eight-complex-column Krylov determinant is

$$-\frac{ix^3}{2^{41}} [x^4 + 7x^3 + (24 + 4i)x^2 + (32 + 26i)x + 24 + 36i].$$

The real and imaginary parts of the bracket have no common root. The cubic zero at  $x = 0$  prohibits reading this exact-rank statement as a uniform sensitivity bound. Neither the weight nor edge selectability has been derived from the primitive source.

### 1.5 A covariant event channel with an actual boundary response

Let  $K$  be the Hermitian one-particle lift of the existing edge  $E_{14}$ . It obeys  $K^3 = K$ . Define

$$U(\theta) = I + (\cos \theta - 1)K^2 - i \sin \theta K, \quad U_k = O^k U(\theta) O^{-k}, \quad \Phi_\theta(\rho) = \frac{1}{6} \sum_{k=0}^5 U_k \rho U_k^*.$$

This is a covariant, trace-preserving completely positive map. Averaging the Hamiltonians first instead preserves the  $P/Q$  split. The two operations must not be confused. For the invariant hidden one-particle state  $Q/3$ , writing  $c = \cos \theta$ ,

$$\text{Tr}(P\Phi_\theta(Q/3)) = \frac{35 - 34c - c^2}{108}.$$

At  $\theta = \pi/2$  the value is  $35/108$ , without retaining an orientation record. The pulse itself acts only on carrier modes. With  $R_B$  the original boundary projector and  $h$  the unchanged source one-body Hamiltonian,  $\rho' = \Phi_{\pi/2}(Q/3)$  obeys

$$R_B \rho' = 0, \quad \text{Tr}(R_B h \rho' h) = \frac{1}{24},$$

and hence

$$\text{Tr}(R_B e^{-ihs} \rho' e^{ihs}) = s^2/24 + O(s^3).$$

The initial hidden state and the vacuum have identical accessible vacuum marginals, but this stipulated protocol separates them. The vacuum remains vacuum because the lifted operations preserve particle number. This is a conditional positive readout result, not a source-derived preparation or implementation. Covariance of density matrices also does not establish a stronger Gauss-law physical-state constraint.

### 1.6 New continuation: transfer succeeds but state selection fails

A possible, explicitly added continuous event model is

$$\mathcal{L}(X) = -i[h, X] + r(\Phi_{\pi/2}(X) - X), \quad r > 0.$$

The event-only semigroup is the Poisson mixture of  $\Phi^n$ ; the rate  $r$  is unselected. With the Hilbert–Schmidt inner product,

$$\text{Re}\langle X, (\Phi - I)X \rangle = -\frac{1}{12} \sum_k \| [U_k, X] \|_{\text{HS}}^2.$$

The Hamiltonian part has zero real quadratic form. Consequently the fixed space of  $\mathcal{L}$  is the common commutant of all  $U_k$  and  $h$ . The three distinct eigenvalues of each  $U_k$  make its commutant equal to that of  $K_k$ .

Exact linear equations give dimensions ten for the edge-only commutant and five after adding  $h$ . The joint algebra is

$$\mathbb{C}I_6 \oplus M_2(\mathbb{C}).$$

Indeed the boundary plane whose three coefficients sum to zero has a rank-two projector  $R$  with  $K_k R = 0$  and  $hR = R$ . The two normalized positive states  $R/2$  and  $(I - R)/6$  are stationary and distinct. The explicit five fixed directions exhaust the computed commutant.

Thus the event mechanism transmits a signal but does not select a unique one-particle state. Full Fock number conservation adds further obstructions. Its 64-dimensional channel superoperator is not Hilbert–Schmidt-selfadjoint, so it is not the strictly positive selfadjoint  $T$  in the logarithmic transfer theorem. This is a typed exclusion, not a claim that no different source-derived transfer or detailed-balance representation can work.

## 1.7 Arithmetic bridges, acceptance gates and provenance

The archived 10 September rotor synthesis reports an exact change to residue/carry coordinates retaining the declared parent dynamics, arithmetic covering operations, conditional logarithmic weights, and finite-energy approximations with explicit error costs. Discarding the carry loses electric dynamics. Norm time, electric time and modular time are not interchangeable. A finite factoring example and static E8 counts do not supply an asymptotic factoring algorithm or a signed global RH proof. No fresh RH-index closure or Hylæan implementation audit was performed in this synthesis.

| Gate | Remaining physical obligation                                              |
|------|----------------------------------------------------------------------------|
| T1   | Primitive choices, elementary operations and quantitative measure.         |
| T2   | Actual half-charged marked E8 seam field, energy/adjoints and scaling.     |
| T3   | One selected local or quasi-local unitary $3 + 1D$ parent.                 |
| T4   | Chiral SM gauge/Weyl construction, anomalies and mirror control.           |
| T5   | Interacting physical continuum and required Lorentz/scattering properties. |
| T6   | Internal gauge couplings and complete neutrino texture and scale.          |
| T7   | Massless quantum spin two and universal coupling from the same parent.     |
| T8   | Selected physical state and a common source functional for readouts.       |

The next falsifiable source gate is to derive a weighted, phase-preserving grammar of elementary events, specify whether its output is a Hamiltonian, a positive transfer or a channel, and generate the appropriate state data without importing the desired  $H_{70}$ , its eigenstates or target couplings. Reconstruction, physical access, robustness and the downstream continuum bridges must then be checked separately.

The local verification results in normal and optimized modes are:

- `source_edge_access.py`: 44 exact checks;
- `covariant_edge_channel.py`: 20 exact checks;
- `edge_channel_stationary.py`: 15 exact checks.

They are not 79 closed physics claims. The complete local provenance and task report archive are indexed in `docs/TFPT_UNIVERSALRAUM_SYNTHESIS_2026-09-12.md`; the nontechnical version is `docs/TFPT_UNIVERSALRAUM_EINFACH_2026-09-12.md`. Earlier local branches retain their own hypotheses and verification records.

## 1.8 Proposed follow-ups and discriminating research directions

These are proposed investigations, not newly completed results or launched worker tasks. Their common acceptance rule is a forward derivation without importing the desired Hamiltonian, its eigenstates or target couplings. A genuinely independently derived primitive action may itself be the Hamiltonian; it need not be artificially required to be a different object.

**F0: Independent reproduction.** Obtain and pin the original 17- and 66-family certificates before using their strongest claims as local premises; then audit the non-Slater and cyclotomic irreducibility certificates. Keep the 133-dimensional product extension separate. Success means the same exact ranks, coefficient fields and class definitions, with numerical conditioning reported separately. Unavailable originals mean “not reproduced”, not “false”.

**F1: Freeze the elementary grammar.** For one original finite sector, map each primitive rule to its operator, phase, support, adjoint and composition. Distinguish physical operations from gauge identifications and elementary generators from composite words. A source-derived filtration or equivalent primitive rule must explain why composition does not automatically introduce independently weighted energy terms. Reject a cutoff chosen only to enforce the desired covariance rank.

**F2: Derive process type and measure.** Construct one source step or collar without the target generator as input. Declare whether the output is a Hamiltonian, a positive Hilbert-space transfer or a reduced quantum channel. Derive weights, step duration and any traced environment from the stated inputs. Test alternative weights with identical symmetry and fixed vector. A freely chosen metric in  $d^*Wd$ , pulse angle or event rate remains an additional datum. Channel positivity does not by itself license a Hamiltonian or Markov logarithm.

**F3: State selection and correlation content.** Compute the exact fixed algebra or ground space of the F2 candidate. Use eigenstate covariance, thermal stationarity or full two-particle data only under their respective contracts. Test identical one-body data with different four-point responses, and identical ground rays with different energy ratios. Repetition of the current channel and free flow cannot remove their common fixed algebra. Any operation removing that algebra must be independently supplied by the source, not by a target-state penalty.

**F4: Operational response and costs.** For a source-derived intervention, compare admissible preparations with the same initial accessible marginal at the original boundary. Bound the remainder in the conditional  $s^2/24 + O(s^3)$  signal on an explicit finite time interval, and include preparation energy, precision and sampling costs. Preserve the no-intervention control. Exact cyclic rank alone is not successful physical control, and arbitrarily small edge asymmetry must not be compensated by unaccounted unlimited precision.

**F5: Source-to-model maps.** An explicit map between the successful candidate and H70, the rotor parent or the charged seam must preserve the relevant adjunction, charge/Clock action, states, dynamics and readouts on named domains. Matching spectra, dimensions or group names alone is insufficient. An open chain must not be identified with a winding rotor, nor a free source with an interacting state without the additional operation being derived.

### Four exploratory hypotheses, not established mechanisms.

1. *Positive frustrated selection:* derive the unavoidable positive minimum and relative defect weights from the source. The failed equal-weight H70 selector is not revived by adjusting its weights to the target.

2. *A genuine collar transfer*: derive a positive transfer directly from the actual boundary construction and its composition. Reflection positivity must not be inferred merely from a positive Gram matrix.
3. *An internal relational reference*: look for an already present source interaction coupling the untouched boundary plane while respecting the correct joint symmetry. Appending an arbitrary environment or projector does not establish source provenance.
4. *Thermal or multi-state readout*: exploit additional independently obtained state responses for robust reconstruction. Optimizing a synthetic temperature for conditioning does not select a cosmic temperature.

**F6: Separate downstream acceptance.** The half-charge, common  $3 + 1D$  parent, chiral, continuum, coupling and gravity obligations remain separate after any finite success. No new RH or factoring mechanism is proposed in this roadmap; future arithmetic work must first consult the existing source-bound proof and counterexample registers. A later Hylæan application could test order-independent consolidation on frozen inputs while preserving distinguishability and recall, but requires an actual implementation map and separate authorization.

F0 and F1 can proceed independently. F2 then supplies a candidate for both F3 and F4; F5 requires a successful typed source result. The immediate recommendation is to freeze one original-sector grammar and reproduce only the external certificates needed for its first forward test. Full acceptance and stop conditions are in `docs/TFPT_UNIVERSALRAUM_FOLLOWUPS_2026-09-12.md`.